

IPC144 – Introduction to C

Programming

Practice Exercises

Week 3

Q1 - (Gas Mileage) Drivers are concerned with the mileage obtained by their automobiles. One driver has kept track of several tankfuls of gasoline by recording miles driven and gallons used for each tankful. Develop a program that will input the miles driven and gallons used for each tankful. The program should calculate and display the miles per gallon obtained for each tankful. After processing all input information, the program should calculate and print the combined miles per gallon obtained for all tankfuls.

Q2 - (Credit-Limit Calculator) Develop a C program that will determine whether a department store customer has exceeded the credit limit on a charge account. For each customer, the following facts are available:

- a) Account number
- b) Balance at the beginning of the month
- c) Total of all items charged by this customer this month
- d) Total of all credits applied to this customer's account this month
- e) Allowed credit limit

The program should input each fact, calculate the new balance (= beginning balance + charges – credits) and determine whether the new balance exceeds the customer's credit limit. For those customers whose credit limit is exceeded, the program should display the customer's account number, credit limit, new balance and the message "Credit limit exceeded."

Q3 - (Sales-Commission Calculator) One large chemical company pays its salespeople on a commission basis. The salespeople receive \$200 per week plus 9% of their gross sales for that week. For example, a salesperson who sells \$5000 worth of chemicals in a week receives \$200 plus 9% of \$5000, or a total of \$650. Develop a program that will input each salesperson's gross sales for last week and will calculate and display that salesperson's earnings. Process one salesperson's figures at a time.

Q4 - (Interest Calculator) The simple interest on a loan is calculated by the formula

$$\text{interest} = \text{principal} * \text{rate} * \text{days} / 365;$$

The preceding formula assumes that rate is the annual interest rate, and therefore includes the division by 365 (days). Develop a program that will input principal, rate and days for several loans, and will calculate and display the simple interest for each loan, using the preceding formula.

Q5 - (Salary Calculator) Develop a program that will determine the gross pay for each of

several employees. The company pays “straight time” for the first 40 hours worked by each employee and pays “time-and-a-half” for all hours worked in excess of 40 hours. You’re given a list of the employees of the company, the number of hours each employee worked last week and the hourly rate of each employee. Your program should input this information for each employee and should determine and display the employee's gross pay.

Q6 - (Factorial) The factorial of a nonnegative integer n is written $n!$ (pronounced “n factorial”) and is defined as follows:

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1 \text{ (for values of } n \text{ greater than or equal to } 1)$$

and

$$n! = 1 \text{ (for } n = 0).$$

For example, $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$, which is 120.

- a) Write a program that reads a nonnegative integer and computes and prints its factorial.
- b) Write a program that estimates the value of the mathematical constant e by using the formula:

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$$

- c) Write a program that computes the value of e^x by using the formula

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$